

Updated aims & scope

What we publish

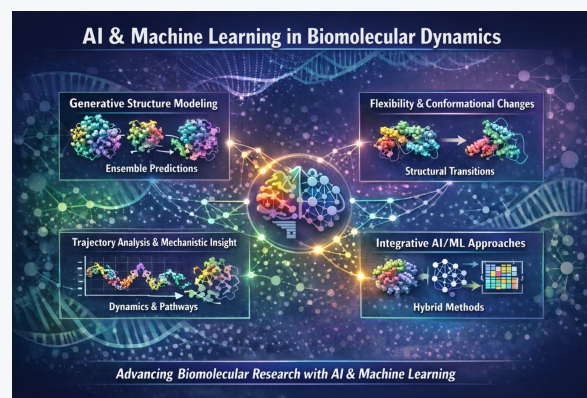
JBSD publishes high-quality experimental, computational, and theoretical biophysical studies that connect biomolecular structure, dynamics, and function.

Examples within scope

- New theoretical, experimental, or computational methods for studying biomolecular structure and dynamics.
- Structural biology: cryo-EM (including time-resolved), NMR, and X-ray crystallography of proteins, nucleic acids, and complexes.
- Molecular dynamics and rigorous enhanced sampling (e.g., metadynamics, accelerated/replica-exchange MD, umbrella or steered MD) for conformational transitions and free-energy landscapes.
- Single-molecule and time-resolved spectroscopies; EPR; and quantitative interaction thermodynamics (e.g., SPR, ITC).
- Hybrid and multiscale modeling, including QM/MM and coarse-grained approaches; and AI-based models for biomolecular structure/dynamics.
- Drug-discovery or docking studies are considered when supported by experiments and when they provide significant biophysical insight.
- Review articles require prior editorial approval; unsolicited reviews are not considered.

Special collection: AI/ML

Generative AI, AI, and Machine Learning in biomolecular structure & dynamics



Accepting submissions through at least all of 2026. Submit via the standard JBSD portal and indicate in your cover letter that you would like the manuscript considered for this special collection.

Suggested topics

- Generative structure modeling; ensemble prediction;
- Learning conformational transitions, allostery, and rare events from simulation and experiment.
- Trajectory analysis for state discovery, kinetics, pathways, and mechanistic interpretation.
- Hybrid physics–ML methods: ML potentials, surrogate models, enhanced sampling, and integrative pipelines.
- Benchmarking, validation, and best practices for AI/ML in biomolecular simulation and molecular biophysics or structural biology



Journal site



Submission
portal



AI/ML collection